NLP Lab Programs – Short Exam Answers

1. Text Preprocessing & Tokenization

```
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

text = "The students are playing football"

words = nltk.word_tokenize(text)
words = [w for w in words if w.isalpha()]

stop = set(stopwords.words("english"))
words = [w for w in words if w.lower() not in stop]

stem = PorterStemmer()
words = [stem.stem(w) for w in words]

print(words)
```

2. Word Classification using Heuristic Rules

```
words = ["playing", "walked", "books", "running"]

for w in words:
    if w.endswith("ing"):
        print(w, "Verb")
    elif w.endswith("ed"):
        print(w, "Past Verb")
    elif w.endswith("s"):
        print(w, "Plural")
```

3. Noun, Verb and Adjective using Dictionary

```
sentence = "smart boy plays football"

noun = ["boy", "football"]
verb = ["plays", "runs"]
adj = ["smart", "big"]

for w in sentence.split():
    if w in noun:
        print(w, "Noun")
    elif w in verb:
        print(w, "Verb")
```

```
    elif w in adj:
        print(w, "Adjective")
```

4. Regular Expressions & Pattern Matching

```
import re

text = "Mail a@gmail.com Visit https://abc.com Date 13/09/2026 Number 25"

print("Email:", re.findall(r'\S+@\S+', text))
print("URL:", re.findall(r'https?://\S+', text))
print("Number:", re.findall(r'\d+', text))
print("Date:", re.findall(r'\d{2}/\d{2}/\d{4}', text))

text = re.sub(r'\d+', '', text)
text = re.sub(r'\s+', ' ', text)
print(text)
```

5. Stemming & Lemmatization Comparison

```
import spacy
from nltk.stem import PorterStemmer, SnowballStemmer

words = ["playing", "studies", "cars"]

p = PorterStemmer()
s = SnowballStemmer("english")
nlp = spacy.load("en_core_web_sm")

for w in words:
    print(w, p.stem(w), s.stem(w), nlp(w)[0].lemma_)
```

Required Installation

requirements.txt:

```
nltk
spacy
requests
```

Install:

```
pip install -r requirements.txt
python -m spacy download en_core_web_sm
```

NLTK data (run once):

```
import nltk
nltk.download("punkt_tab")
nltk.download("stopwords")
```